

SANTANDER X · QUANTUM AI LEAP · 2026

EmpowerTours Quantum Portfolio

Q-Day-resistant DeFi, shipped today.

QAOA on IBM Heron · Hedged post-quantum signatures · On-chain
provenance on Monad

THE THREAT

Every ECDSA signature you publish today is a hostage to tomorrow's QPU.

Harvest now. Decrypt later. The adversary already has your 2026 trade orders — they just haven't broken the curve yet.

WHY DEFI GETS HIT FIRST

Public chains are an open archive of signatures.

- Every transaction broadcasts an **ECDSA signature** in the clear.
- Indexers store them **forever**.
- A cryptographically-relevant QPU forges any one of them.
- The chain doesn't know which.

TODAY'S DEFENSE

None. Most wallets still sign with secp256k1.

Most "post-quantum DeFi" decks are **roadmaps**.

THE STACK

Three layers. Each one survives the other two.

1. ALLOCATE

QAOA on IBM

Heron QPU picks
Markowitz-optimal
weights under risk
constraints.

2. SIGN

Hedged PQ: every
order carries ML-
DSA-65 + SLH-
DSA-SHAKE-256s +
Ed25519. **Any one**
survives.

3. ANCHOR

SHA-256(order)

committed to
Monad's

`AuditAnchorV2`

before the vault
will execute it.

*One layer broken $\neq$ system broken. The threat model is **every algorithm***

THE QUANTUM LAYER

QAOA on real hardware, reported with statistical honesty.

- Executed on **IBM Heron r2** (`ibm_marrakesh`) via Qiskit Runtime, 4 096 shots, raw and error-mitigated (XY4 DD + measurement twirling). Job IDs shipped in `outputs/hardware_run_defi.json`.
- **n = 1 per arm.** We report

WHAT THIS IS, HONESTLY

QAOA is **not** faster than the classical baseline at this size — that baseline is

`NumPyMinimumEigensolver` (brute-force exact, `src/solvers.py:33`), which is optimal and instant on 8 assets. We measure the **gap**

THE SIGNATURE LAYER

Don't bet on one algorithm. Hedge the bet.

ALGORITHM	FAMILY	WHY INCLUDE
ML - DSA - 65	Lattice (FIPS 204)	NIST PQ standard. Fast.
SLH - DSA - SHAKE - 256s	Hash (FIPS 205)	Different math. Slow but conservative.
Ed25519	Classical EC	Battle-tested. Hedge for "did we mis-port a new standard?"

If ML-DSA falls to a 2028 cryptanalysis paper, SLH-DSA still authenticates the order. If both lattice and hash fall, the classical signature carries it

THE PROVENANCE LAYER

A hash chain the vault refuses to disobey.

1. Agent builds order. PQ-signs it.
2. `AuditAnchor.anchor(orderHash)` on Monad. Links to caller's previous hash.
3. `RoutingVault.executeAndRoute(orderHash, ...)` refuses execution unless `ANCHOR.lastHash[msg.sender] == orderHash`.

WHAT THIS KILLS

- Replay (sequence enforced per wallet)
- Off-chain order tampering (hash doesn't match)
- Vault impersonation (pair

PROOF, NOT PROMISES

Live on Monad MAINNET. 279 tests. ZK-verified.

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CONTRACTS LIVE
MONADSCAN-VERIFIED

279

TESTS PASSING
110 PYTHON + 169
FOUNDRY

3

SIGNATURE ALGORITHMS
PER ORDER, HEDGED

AuditAnchorV2	0x8422b555dce11913a4657c2f47c839637fc71ffd	
UniswapRoutingVault	0x06f233062ee23590e5cc873df511024f3d981e56	
MorphoSupplyAdapter	0x8d5ae2f23e5d20bfb7915168d6b2a3ce753fe49e	
MLDSAAttestation	0xb0aadafe68647578520e988b4444e556c300b4da	

THE END-TO-END DEMO — REAL VALUE, ONE HASH

QPU decision → PQ signature ZK-verified on-chain → yield.

STEP	CONTRACT	WHAT HAPPENED (MONAD MAINNET)
1. Attest	MLDSAAttestation	Order's ML-DSA-65 signature verified on-chain via ZK proof (~230k gas, not ~500M); <code>pqAttested = true</code>
2. Anchor	AuditAnchorV2	Hash <code>0xd8bf1551...15f9</code> committed, immutable
3. Swap	UniswapRoutingVault	0.1 MON → 2,123 micro-USDC (0.002123 USDC ≈ \$0.002 — a deliberately tiny live-value demo) via live Uniswap v3, anchor-gated
4. Yield	MorphoSupplyAdapter	USDC supplied into a live Morpho market (~4.75% APY), non-custodial

WHY US

We ship the part everyone else handwaves.

- **Most PQ-DeFi pitches:** slides about a future migration.
- **Most QAOA-finance pitches:** one hand-picked seed on a simulator.
- **Most provenance pitches:** off-chain Merkle trees nobody verifies.
- **Ours:** verified contracts, real QPU runs with statistical rigor, on-chain hash chain enforced by the vault itself.
- One repo. One CI. Reproducible from a cold clone.

WHAT'S NEXT

From live mainnet proof to institutional pilot.

Now → Q3 2026

- Independent third-party security audit (contracts + PQ/ZK)
- Package as a B2B SDK (wallets, custodians, bridges)
- HSM/KMS-backed key custody

Q4 2026

- First institutional design-partner pilot
- Hot-path PQ cosigner service
- Paid bug bounty (Immunefi / Code4rena)

THE ASK

Pilot with Santander.

**Q-Day-resistant crypto exposure for institutional treasury,
on rails we can prove are honest.**

`github.com/EmpowerTours/quantum-portfolio · commit 33b69dd`